

# STD strength matching between one and two depression timescales (2026-09-13)

*Decision record for the paper's single- vs multiple-timescale comparison. Equations and the authoritative statement: docs/EquationsParametersDocs/Equations\_stability\_paper.md, § "2026-09-13: STD strength matching". Presets: srnn\_param\_presetcases celltype\_pairs\_sfaEI\_Sc0p2sig0p1\_noise0p025\_dualStdScaled\_3cond\_mu8p25 and ...\_dualStdUsage\_3cond\_mu8p25. Test: scripts/tests/test\_route\_scale.m. Steady-state curves over the occupied rates: fig\_STD\_steady\_state.*

## 1. The decision

**Made by TR, 2026-09-13**, on the plan for the overnight of 2026-09-13/14.

The paper compares no adaptation, one-timescale adaptation (1 SFA + 1 STD) and multiple-timescale adaptation (3 SFA + 2 STD) and attributes the differences to the *temporal structure* of adaptation. SFA is normalised for that comparison:  $c_{\text{eff}} = c/K$  makes the steady-state adaptation  $c \cdot r$  whatever  $K$  is. STD was not. With both depression timescales at  $\rho = \tau_{\text{rel}}/\tau_{\text{rec}} = 0.125$  ( $\tau_{\text{rec}} [2\ 4]$  s,  $\tau_{\text{rel}} [0.25\ 0.5]$  s) and the single-timescale routes at (2, 0.25), the two-timescale steady state  $1/(1 + r/\rho)^2$  is the **square** of the one-timescale  $1/(1 + r/\rho)$ , so adding a timescale also multiplied the depression. Codex's audit (scripts/paper/fig\_to\_do\_in\_future.md § 1) and StochasticPlasticDynamicalSystemPaper/reports/suggest\_updates\_main\_matlab\_model.md both flagged this as the model-defining decision to freeze before any final run.

**What is matched:** the steady-state synaptic output  $\theta_{\text{ss}}(r) = r \cdot \Pi_m b_m(r)$  of the two-timescale routes equals that of the one-timescale routes at a reference rate  $r_{\text{ref}}$ , chosen by rule as the occupied median rate (below).

**How (primary): a multiplicative scale on the two-timescale routes.** The depression variables,  $\theta$  and the ODE are untouched; the weights of the four sfa3\_std2 routes are multiplied by

$$s = \theta_{\text{single}}(r_{\text{ref}}) / \theta_{\text{dual}}(r_{\text{ref}}) = 1 + r_{\text{ref}}/\rho = 3.$$

Implemented as `synapse_config.<pre>.<post>.scale`, applied to `params.W` in `SRNNCellTypePairs.get_params` only (the drawn  $W$  stays unscaled; dynamics, both Jacobians and `jacobian_times` read `params.W`, so they agree by construction; the ESN 'synaptic' readout's route-redundancy check compares the scale too). `no_adaptation` and `sfa1_std1` are bit-identical to the unmatched preset's.

**Direction:** scale the dual condition *up* to the single one ( $s = 3$  on dual, 1 on single), so the single-timescale condition stays exactly the current network. Alternatives offered and not taken: single scaled down to dual ( $s \approx 0.33$ ), or both adapted conditions scaled to no adaptation ( $s = 3$  and 9).

**Control (also run):** the usage-matched preset, which needs no code change.

## 2. The alternative that was raised and overruled

The recommendation put to TR was to match by **usage**: keep  $\tau_{\text{rec}} = [2\ 4]$  s and choose  $\tau_{\text{rel}}$  so the product equals the single factor at  $r_{\text{ref}}$ . With equal usage  $\rho_u$  on both timescales,

$$(1 + r_{\text{ref}}/\rho_u)^2 = 1 + r_{\text{ref}}/\rho \rightarrow \rho_u = r_{\text{ref}} / (\sqrt{1 + r_{\text{ref}}/\rho} - 1) = 0.25/(\sqrt{3} - 1) = 0.34151$$
$$\tau_{\text{rel}} = \rho_u \cdot \tau_{\text{rec}} = [0.68301, 1.36603] \text{ s}.$$

Reasons it was recommended:  $\tau_{\text{rel}}$  is Varela's release strength  $d_m$ , so this is the tuning the product model itself offers; it changes no code; and it leaves the **undepressed (low-rate) recurrent gain identical across the three conditions**. Its cost: each timescale depresses less and the depression dynamics are slower (rate  $1/\tau_{\text{rec}} + r/\tau_{\text{rel}}$  per variable).

The concern with the scale, stated to TR before the choice: because  $b_m \rightarrow 1$  as  $r \rightarrow 0$ , the scaled dual condition runs at **3× the undepressed recurrent gain** of the other two conditions. The match holds at the operating point, not at low rates, and a change in effective  $W$  scale is itself a change the paper's `level_of_chaos` sweep treats as a stability axis. TR reaffirmed the scale as primary and asked for the usage variant as the control, so the morning comparison (§ 5) shows the consequence rather than arguing it.

## 2b. The third direction, built after the fast runs (2026-09-14, 03:15)

Both matchings above make the two-timescale routes WEAKER. Their fast runs (§ 5) turned the multiple-timescale network chaotic and removed its memory advantage, so a third preset was added that leaves the two-timescale routes exactly as published and instead STRENGTHENS the single-timescale route to the dual's steady state at  $r_{\text{ref}}$ :

$$1/(1 + r_{\text{ref}}/\rho_s) = 1/(1 + r_{\text{ref}}/\rho)^2 = 1/9 \quad \text{at } r_{\text{ref}} = 0.25, \rho = 0.125$$

$$\rho_s = r_{\text{ref}} / 8 = 0.03125$$

$$\tau_{\text{rel}} = \rho_s * \tau_{\text{rec}} = 0.03125 * 2 \text{ s} = 0.0625 \text{ s} \quad (\text{was } 0.25 \text{ s})$$

celltype\_pairs\_sfaEI\_Sc0p2sig0p1\_noise0p025\_dualStdSingleMatched\_3cond\_mu8p25: no\_adaptation and sfa3\_std2 are byte-identical to the unmatched preset; only sfa1\_std1 changes ( $\tau_{\text{rec}}$  2 s kept,  $\tau_{\text{rel}}$  0.25 -> 0.0625 s, i.e. the usage  $d = 1/\tau_{\text{rel}}$  quadrupled). No route scale, no model code. Consequences: the single route's steady state is monotone in  $r$  and saturates at  $\rho_s$ , so it matches the dual product only at  $r_{\text{ref}}$  (the dual peaks near  $r = \rho$  and falls); its relaxation rate  $1/\tau_{\text{rec}} + r/\tau_{\text{rel}}$  at  $r_{\text{ref}}$  rises from 1.5 to 4.5  $\text{s}^{-1}$ , so the single-timescale condition depresses both more strongly and faster than before. The undepressed gain is unchanged in every condition.

Three matchings were therefore tried, in this order: (i) route scale  $s = 3$  on the dual routes (TR's choice), (ii) equal usage  $\rho_u = 0.342$  on both dual timescales (the control), (iii) single route strengthened, dual as published. (i) and (ii) equalise by weakening the dual; (iii) by strengthening the single.

## 3. The numbers

**$r_{\text{ref}} = 0.25$ .** Rule: the median mean firing rate of the multiple-timescale condition at the default point of the 1-D sweeps of the medium run data/topk\_med (top-K, 15 reps per level), rounded to 0.05. Per sweep, the level nearest the preset default:

| sweep          | sfa3_std2 median rate |
|----------------|-----------------------|
| f_E            | 0.236                 |
| level_of_chaos | 0.223                 |
| mu_EE_relative | 0.239                 |
| mu_IE_relative | 0.251                 |
| n              | 0.231                 |

( $\mu_{\text{EI\_relative}}$  and  $\mu_{\text{II\_relative}}$  were excluded: their default is negative and the nearest-level lookup landed on the wrong sign.) Pooled over the 105 near-default reps:

| condition     | median | p5    | p25   | p75   | p95   |
|---------------|--------|-------|-------|-------|-------|
| no_adaptation | 0.483  | 0.027 | 0.274 | 0.729 | 0.915 |
| sfa1_std1     | 0.269  | 0.033 | 0.167 | 0.432 | 0.791 |
| sfa3_std2     | 0.237  | 0.050 | 0.220 | 0.249 | 0.346 |

The joint 64-point parameter-space sample, which ranges far from the default, has medians 0.250 / 0.159 / 0.154 for the same three conditions; it was not used because it is not the operating point.

**At  $r_{\text{ref}} = 0.25$ ,  $\rho = 0.125$ :** one factor  $1/(1 + 2) = 1/3$ ; unmatched product  $1/9$ ; scale  $s = 3$ ; usage  $\rho_u = 0.34151$ ,  $\tau_{\text{rel}} = [0.68301, 1.36603] \text{ s}$ . Both matched variants give  $\theta_{\text{ss}}(0.25) = 0.25/3 = 0.0833$ , against 0.0278 unmatched.

| quantity at $r_{\text{ref}}$                   | single (1 TS) | dual unmatched | dual scaled | dual usage |
|------------------------------------------------|---------------|----------------|-------------|------------|
| $\Pi b_m$                                      | 0.333         | 0.111          | 0.111       | 0.333      |
| $\theta_{\text{ss}} = s \cdot r \cdot \Pi b_m$ | 0.0833        | 0.0278         | 0.0833      | 0.0833     |
| undepressed gain factor ( $r \rightarrow 0$ )  | 1             | 1              | 3           | 1          |

## 4. Replacement manuscript text

The manuscript repo is not edited here; this is ready to paste into Manuscript5.md.

### 4a. What it says now (Supplemental Methods, “Short-term depression”, ~line 375)

Unlike adaptation, depression is deliberately not normalized by its number of timescales: it enters as a product rather than a sum, each factor rests at one, and adding a timescale is meant to deepen depression rather than redistribute it, following the product-of-components description of cortical depression [varelaQuantitativeDescriptionShortTerm1997].

and, earlier in the same paragraph, “so the product in  $\{\text{eq:theta}\}$  settles to  $(1 + 8r)^{-2}$ . At  $r = 1$  a single timescale reduces synaptic output ninefold whereas two reduce it by a factor of 81, the square  $\{\text{supFig:std\_steady\_state}\}$ .”

### 4b. Replacement paragraph (scaled variant primary, usage control)

**Short-term depression.** The same depression time constants were applied to all four connection routes. In the single-timescale condition the recovery time constant was eight times the release time constant, so at a constant rate  $r$  the depression variable settles to  $b = 1/(1 + 8r)$ . Depression enters  $\{\text{eq:theta}\}$  as a product of resource variables, each resting at one, following the product-of-components description of cortical depression [varelaQuantitativeDescriptionShortTerm1997]; with the same ratio on both timescales the two-timescale product would settle to  $(1 + 8r)^{-2}$ , the square of the single-timescale value. Adding a depression timescale would then change the strength of depression as well as its temporal structure, and the comparison between the adapting conditions would not isolate the number of timescales. We therefore matched the steady-state synaptic output of the two-timescale routes to that of the single-timescale routes at a reference rate  $r_{ref} = 0.25$ , the median mean firing rate of the multiple-timescale network at the reference parameters. In the primary configuration the two-timescale routes retained their release and recovery time constants and their weights were multiplied by a route scale  $s = \theta^{(1)}(r_{ref})/\theta^{(2)}(r_{ref}) = 1 + 8r_{ref} = 3$ , so that the recurrent input to neuron  $i$  is  $\sum_j s_{q_j \rightarrow q_i} w_{ij} \theta_j$  with  $s = 3$  on all four routes of the multiple-timescale condition and  $s = 1$  otherwise. This equalises synaptic output at the operating point; because the resource variables approach one at low rates, the undepressed recurrent gain of the multiple-timescale condition is three times that of the other conditions. As a control we also ran a usage-matched configuration in which the weights were unchanged and the release time constants of the two depression variables were lengthened to  $\tau_{rel} = 0.683$  and  $1.366$  s, giving both variables the usage ratio  $\tau_{rel}/\tau_{rec} = 0.342$  for which  $(1 + r_{ref}\tau_{rec}/\tau_{rel})^2 = 1 + 8r_{ref}$ . The two configurations agree at  $r_{ref}$  and differ away from it  $\{\text{supFig:std\_steady\_state}\}$ . The single-timescale and no-adaptation conditions are identical in both. Where the text attributes a difference between the adapting conditions to the number of adaptation timescales, it refers to this strength-matched comparison; the earlier unmatched configuration compared timescale count and depression strength together.

### 4c. Replacement rows for

| Condition                                   | $\tau_a$ (s)    | $\tau_{rel}$ (s) | $\tau_{rec}$ (s) | route scale $s$ |
|---------------------------------------------|-----------------|------------------|------------------|-----------------|
| No Adaptation                               | none            | none             | none             | 1               |
| Single-Timescale Adaptation                 | 0.25            | 0.25             | 2                | 1               |
| Multiple-Timescale Adaptation (primary)     | 0.25, 1.581, 10 | 0.25, 0.5        | 2, 4             | 3               |
| Multiple-Timescale, usage-matched (control) | 0.25, 1.581, 10 | 0.683, 1.366     | 2, 4             | 1               |

Caption addition: “The multiple-timescale routes are strength-matched to the single-timescale routes at  $r_{ref} = 0.25$ : in the primary configuration by a route weight scale  $s = 3$ , in the control by the release time constants.”

The parameters table rows for  $\tau_{rel}$  (E to E ... I to I) stay  $[0.25 \ 0.5]$  for the primary configuration; add a row “route scale  $s$  (all four routes, multiple-timescale condition) | STD strength match | 3 | -”.

## 5. Results (fast runs of all three matchings, 2026-09-14)

Near-default = the seven 1-D sweeps at the level nearest the preset default, pooled (21 networks per condition at fast; 105 in the medium unmatched run). Memory capacity: total MC over 15 s of delays, 5 paired trials at fast, 15 at medium. The no-adaptation row is identical physics in every column and the single-timescale row is identical in the scaled and usage columns; the small differences between the unmatched and the fast columns there are the run mode (fewer reps, shorter T), which is the calibration for reading the rest.

| measure                                                     | unmatched<br>(data/topk_med,<br>medium) | (i) scaled s = 3<br>(data/stdscaled_fast) | (ii) usage rho_u =<br>0.342<br>(data/stdusage_fast) | (iii) single<br>strengthened<br>(data/stdsinglematched_fast)                                                                      |
|-------------------------------------------------------------|-----------------------------------------|-------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| lambda_1 near<br>default, none /<br>single / multiple       | +3.47 / +0.35 /<br><b>-0.112</b>        | +3.65 / +0.61 /<br><b>+2.24</b>           | +3.65 / +0.61 /<br><b>+1.01</b>                     | +3.65 / (row<br>missing from the<br>figure table, to<br>check: its<br>joint-sample median<br>is -0.51) / <b>-0.118</b><br>15 / 21 |
| networks with<br>lambda_1 < 0,<br>multiple                  | 105 / 105                               | 0 / 21                                    | 0 / 21                                              | 15 / 21                                                                                                                           |
| occupied median<br>rate, none / single /<br>multiple        | 0.483 / 0.269 /<br>0.230                | 0.429 / 0.215 /<br>0.300                  | 0.429 / 0.215 /<br>0.276                            | 0.429 / 0.211 /<br>0.236                                                                                                          |
| K the top-K needed<br>for the<br>multiple-timescale<br>jobs | 15                                      | 60 (D_KY resolved<br>0-50%)               | 15-60 (39-100%)                                     | 15 (67-100%)                                                                                                                      |
| tau sweep,<br>multiple-timescale,<br>slowest tau 1 -> 30 s  | -0.112 -> -0.043, all<br>negative       | +2.3 flat, all positive                   | +0.65 to +1.14, all<br>positive                     | (figure pending; K =<br>15 at every level)                                                                                        |
| total MC, none /<br>single / multiple                       | 0.103 / 0.259 /<br><b>0.590</b>         | 0.111 / 0.243 /<br><b>0.103</b>           | 0.111 / 0.243 /<br><b>0.146</b>                     | 0.111 / <b>0.385</b> /<br><b>0.583</b>                                                                                            |
| MC horizon (s)                                              | 0.00 / 0.12 / 0.52                      | 0 / 0.18 / 0                              | 0 / 0.18 / 0                                        | 0 / 0.30 / 0.48                                                                                                                   |
| single vs multiple<br>MC                                    | p = 1.2e-4, d_z =<br>-2.04              | p = 0.0625, d_z =<br>+1.66 (wrong way)    | p = 0.0625, d_z =<br>+1.24 (wrong way)              | p = 0.125 (floor<br>0.0625), d_z = -1.05                                                                                          |

Reading. (i) and (ii): at equal steady-state depression obtained by weakening the dual routes, the multiple-timescale network is chaotic and has no fading memory; the unmatched advantage leaned on the squared depression. (iii): with the dual network as published and the single route brought to the same steady state, the multiple-timescale network keeps lambda\_1 ~ -0.12 and MC ~ 0.58, while the single-timescale network improves (MC 0.24 -> 0.39, horizon 0.12 -> 0.30). Both mechanisms contribute: depression strength moves the single condition part of the way, and two timescales at the same strength go the rest. The claim that survives is “at matched steady-state depression, two depression timescales extend fading memory further than one”, with a smaller effect size than the unmatched comparison showed.

The occupied median rate of the multiple-timescale condition stays within 0.23-0.30 in every variant, so r\_ref = 0.25 does not need revisiting. fig\_STD\_strength\_matching (in each run's figs/<run>/) has theta\_ss at r\_ref and at the occupied percentiles and the max |log ratio| over the occupied band: 1.11 unmatched, 0.146 scaled, 0.034 usage (the single-strengthened curve can be drawn by passing that preset as reference\_preset).

h\_KS and D\_KY medians per condition are in each run's parameters.md Lyapunov quality table and the Fig\_Sensitivity\_\*\_medians sheets; they were not tabulated here because the K column above already carries the message.

#### 4d. Replacement paragraph if (iii) becomes the paper preset

**Short-term depression.** Depression enters  $\{\text{eq:theta}\}$  as a product of resource variables, each resting at one, following the product-of-components description of cortical depression []. In the multiple-timescale condition the two depression variables had recovery time constants of 2 and 4 s and release time constants of 0.25 and 0.5 s, so that at a constant rate settled to  $\frac{1}{91}$ . The numbers  $\frac{1}{(1 + 8r)}$  their product to  $\frac{1}{10490891 + 8r} \approx \frac{1}{10490891}$ . A single depression variable with the same ratio would settle to  $\frac{1}{91}$ . The numbers  $\frac{1}{(1 + 8r)}$  so a one-timescale condition built that way would be weaker as well as structurally simpler, and the comparison between the adapting conditions would not isolate the number of timescales. We therefore matched the steady-state synaptic output of the single-timescale routes to that of the two-timescale routes at a reference rate = 0.25 the median mean firing rate of the multiple-timescale network at the reference parameters: the single depression variable kept its recovery time constant of 2 s and its release time constant was shortened to 0.0625 s, for which  $\frac{1}{91}$ . The numbers  $\frac{1}{(1 + r_{\text{ref}})\tau_{\text{rec}}/\tau_{\text{rel}}}$  =  $\frac{1}{(1 + 8r_{\text{ref}})^{-2}} = \frac{1}{91}$ . The two conditions agree at differ away from it ( $\{\text{supFig:std\_steady\_state}\}$ ); the multiple-timescale and no-adaptation conditions are unchanged. Where the text attributes a difference between the adapting conditions to the number of adaptation timescales, it refers to this strength-matched comparison.

Conditions-table row for the single-timescale condition under (iii): 1 SFA ( $\tau_a$  0.25 s) / 1 STD ( $\tau_{\text{rel}}$  0.0625 s,  $\tau_{\text{rec}}$  2 s), all four routes.

#### 6. Open questions for TR (updated 2026-09-14 morning)

1. **Which matching is the paper's.** The fast runs settle the direction question TR was asked on 2026-09-13: (i) scale-up and (ii) usage both destroy the multiple-timescale result; (iii) keeps it. My recommendation is (iii), `..._dualStdSingleMatched_3cond_mu8p25`, run at medium (`clone stdUsage_med_config`). `paper_config` still points at (i) and should be moved with one line.
2. **What the paper claims.** Under (iii) the honest sentence is that two depression timescales at matched strength extend fading memory and keep  $\lambda_1$  near zero where one does not, AND that depression strength itself contributes (the single condition improved when strengthened). The product-of-components model stays primary; the matching is a control on the single-timescale condition, which is the least disruptive option the `suggest_updates` report recommended.
3. **Route scale.** Keep `synapse_config.<pre>.<post>.scale` in the model (it is tested and harmless at 1) or remove it now that no paper preset uses it.
4. **Effect size.** At 5 trials the single-vs-multiple MC test cannot go below  $p = 0.0625$ ; the medium run of (iii) with 15 trials is what the manuscript needs, and the unmatched 15-trial  $p$  of  $1.2e-4$  should not be quoted for the matched comparison.